

$$* |\psi\rangle = \sum_n \alpha_n |\psi_n\rangle$$

$$\langle \psi_m | \psi \rangle = \alpha_m$$

$$\langle x | \psi \rangle = \psi(x)$$

$$\psi(x) = \sum_n \alpha_n \psi_n(x)$$

$\leadsto$

$$|\psi\rangle = \sum_n \alpha_n |\psi_n\rangle$$

$$\langle \psi_m | \psi_n \rangle = \delta_{nm}$$

$$\langle \psi_m | \psi \rangle = \sum_n \alpha_n \langle \psi_m | \psi_n \rangle$$

$$|\psi\rangle = \sum_n \alpha_n |\psi_n\rangle$$

$$= \sum_n \langle \psi_n | \psi \rangle |\psi_n\rangle$$

$$= \left(\sum_n |\psi_n\rangle \langle \psi_n| \right) |\psi\rangle$$

$$\mathbb{1} = \sum_n |\psi_n\rangle \langle \psi_n|$$

$$* \psi(p) = \int d^u x \, e^{i p x} \psi(x)$$

$$|\psi\rangle = \int d^u q \, \alpha(q) |q\rangle$$

$$\langle a | \psi \rangle = \int d^u q \, \alpha(q) \langle a | q \rangle$$

$$\langle a | \psi \rangle = \alpha(a)$$

$$\Rightarrow |\psi\rangle = \int d^u x \, \langle q | \psi \rangle |q\rangle$$

$$= \left(\int d^u x \, |q\rangle \langle q| \right) |\psi\rangle$$

$$\mathbb{1} = \int d^u x \, |q\rangle \langle q|$$